LOTQ-0005 / FUNCTIONS / BEGINNER / FIELD GUIDE

Functions, Graphs, and Transformations

Function behavior, graph transformations, feature maps, and quant pipeline intuition.

Library of the Quant Educational reference manual

Manual Profile

LIBRARY ID	LOTQ-0005
CATEGORY	Functions
DIFFICULTY	Beginner
FORMAT	Single-column field-guide manual
USE	Study, lookup, and research-system reference

Educational Use Only

This manual is educational material for quant research study. It is not financial advice, not a trading recommendation, and not a guarantee of correctness or live trading usefulness.

ORIENTATION

BEGINNER

How to Use This Manual

Reading method

Use each page as a standalone term card. Read the one-sentence meaning first, then the memory jogger, then the quant-use and system-fit boxes. Return to formula and implementation sections when building code.

Study loop

skim concept → say memory jogger → trace example → write formula → map to system → answer recall

For active recall, cover the title and try to reconstruct: what is the input, what is the output, where does it enter a trading research pipeline, and what bug would misuse it create?

Core mental model

Every quant research artifact can be viewed as a function: data validation maps messy raw data to clean tables; feature engineering maps prices to predictors; a model maps features to forecasts; a portfolio rule maps forecasts to positions; a backtest maps positions and prices to diagnostics.

Manual section	How to use it
Meaning + jogger	Fast memory reset.
Quant use	Why the term matters in research.
Example	Concrete trading or market interpretation.
Formula	Mathematical skeleton.
Implementation	How the concept becomes code or system design.
Confusions	Common mistake to avoid.

NAVIGATION

BEGINNER

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NAVIGATION

BEGINNER

Table of Contents continued

FUNCTIONS

Function

BEGINNER

One-sentence meaning

A rule that maps an input to exactly one output.

Memory jogger

Machine that takes x and returns y.

Why it matters in quant research

Use it to express signals, transforms, models, payoff curves, and risk functions as reproducible mappings.

Simple market / trading example

f(x)=x^2 maps 3 to 9; a signal function maps today's features to tomorrow's position.

Formula intuition

f: X -> Y; y=f(x)

Python / system implementation idea

Represent every research step as a pure function when possible: output = transform(input, parameters).

Inline workflow diagram

input → **function** → **output** → **diagnostic** → **decision**

Read the concept as one node in a larger pipeline. The key question is always whether its input is valid and timestamp-safe.

Where this fits in my research system

Feature generator, model layer, scoring function, risk transform.

Confusions to avoid

A function is not just a formula; it can be a table, algorithm, or API call as long as each input has one output.

Related terms

relation, domain, range, mapping

Validation / implementation checklist

Check	What to verify
Input contract	Correct units, allowed domain, no missing values, timestamp-safe availability.
Output contract	Expected shape, range, sign convention, and no impossible values.
Backtest role	Whether the function creates features, targets, signals, positions, costs, or diagnostics.
Robustness	Neighboring parameters and adjacent regimes do not completely reverse the conclusion.

Mini recall prompt

Explain this concept in one sentence, name one formula or code representation, and describe one way it can create a false backtest if misaligned.

System-design connection

Treat this as a node with an explicit input schema, output schema, parameters, version, and timestamp rule. That is what makes research reproducible.

FUNCTIONS

BEGINNER

Input, Output, Domain, Range

One-sentence meaning

Inputs are allowed values; outputs are produced values; domain and range describe their sets.

Memory jogger

Domain is what you may feed it; range is what can come out.

Why it matters in quant research

Bad domains create invalid features, divide-by-zero bugs, lookahead leakage, or nonsense model inputs.

Simple market / trading example

Log return needs positive prices. Price <= 0 is outside the useful domain of ln(P_t/P_{t-1}).

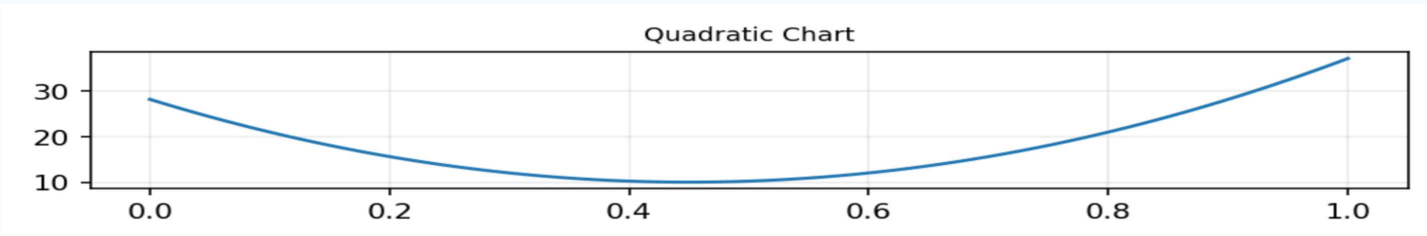
Formula intuition

domain(f) = {x | f(x) is defined}

Python / system implementation idea

Add assert statements for domains before running backtests.

Visual intuition



Synthetic chart for shape intuition; use it as a pattern detector, not as market evidence.

Where this fits in my research system

Data validation, schema checks, feature constraints.

Confusions to avoid

The range is not necessarily every possible output type; it is what the function actually reaches.

Related terms

codomain, support, constraints, null values

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FUNCTIONS

BEGINNER

Independent and Dependent Variables

One-sentence meaning

The independent variable is the input; the dependent variable is the output affected by it.

Memory jogger

x moves first; y responds.

Why it matters in quant research

Helps separate features from targets and avoid mixing predictor variables with future outcomes.

Simple market / trading example

Lookback volatility is independent input; next-day return is dependent target in a supervised model.

Formula intuition

$y = f(x)$

Python / system implementation idea

Store features and targets in separate tables with explicit timestamps.

Failure-mode table

Mistake	Symptom	Fix
Invalid domain	NaN, inf, impossible values	Add validation before transform
Wrong timestamp	Backtest too good	Shift features; audit feature time
Overfit parameter	Only one setting works	Run sensitivity sweep

Where this fits in my research system

Feature matrix X and target vector y.

Confusions to avoid

Dependent does not mean causally caused; it only means modeled as output.

Related terms

feature, label, target, response

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FUNCTIONS

BEGINNER

Function Notation

One-sentence meaning

Notation such as f(x), g(t), or h(r_t) names a mapping and its input.

Memory jogger

The name tells you which machine; the parentheses tell you what you feed it.

Why it matters in quant research

Makes research pipelines explicit: normalize(), rank(), winsorize(), fit(), predict().

Simple market / trading example

signal_t = f(momentum_t, volatility_t, spread_t).

Formula intuition

f(x; theta) where theta are parameters

Python / system implementation idea

Use named function signatures: make_features(prices, lookback=20).

Quick recall check

Name the input, output, valid domain, likely quant use, and one production bug this concept could create if implemented carelessly.

Where this fits in my research system

Pipeline code, model signatures, experiment logs.

Confusions to avoid

f(x) is output, not multiplication. Parameters are settings; variables are inputs.

Related terms

arguments, parameters, signature

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FUNCTIONS

BEGINNER

Graphs of Functions

One-sentence meaning

A graph plots input-output pairs to reveal shape, slope, curvature, and behavior.

Memory jogger

A picture of what the rule does.

Why it matters in quant research

Graphs expose nonlinear transforms, thresholds, saturation, unstable regions, and sensitivity.

Simple market / trading example

Plot transaction cost as a function of trade size to see when costs become nonlinear.

Formula intuition

Graph = {(x, f(x)) | x in domain}

Python / system implementation idea

Plot transforms on synthetic ranges before applying to market data.

Inline workflow diagram

input → function → output → diagnostic → decision

Read the concept as one node in a larger pipeline. The key question is always whether its input is valid and timestamp-safe.

Where this fits in my research system

Diagnostics, feature sanity checks, model interpretability.

Confusions to avoid

A graph is not proof; it is a diagnostic view of sampled points.

Related terms

plot, curve, axes, scale

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FUNCTIONS

BEGINNER

Linear Function

One-sentence meaning

A function with constant slope; output changes by the same amount for each input step.

Memory jogger

Straight line, steady rate.

Why it matters in quant research

Baseline for factor models, exposure estimates, slippage approximations, and first-pass sanity checks.

Simple market / trading example

Expected return = alpha + beta * market_return.

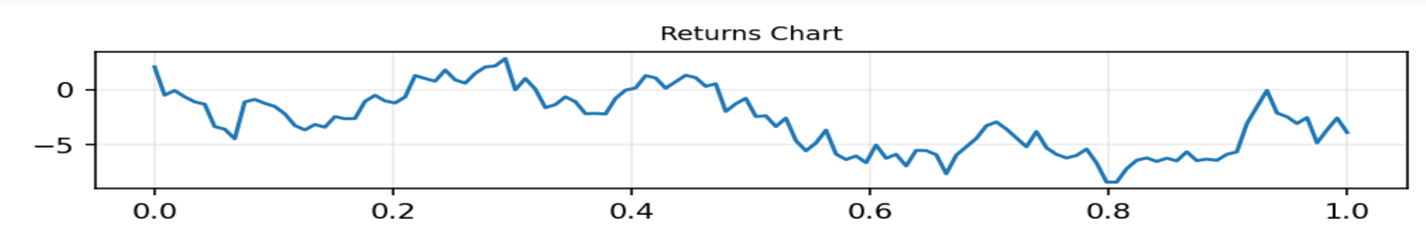
Formula intuition

y = mx + b

Python / system implementation idea

Use LinearRegression only after checking residual structure and time splits.

Visual intuition



Synthetic chart for shape intuition; use it as a pattern detector, not as market evidence.

Where this fits in my research system

Factor model, regression baseline, risk exposure.

Confusions to avoid

Linear in variables is different from linear in parameters.

Related terms

slope, intercept, beta, affine

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FUNCTIONS

BEGINNER

Slope / Rate of Change

One-sentence meaning
Slope measures output change per unit input change.

Memory jogger
How much y moves when x moves one notch.

Why it matters in quant research
Core intuition behind beta, sensitivity, marginal risk, delta, and gradient-based optimization.

Simple market / trading example
If PnL changes $2,000 for a 1% market move, slope is exposure to that market move.

Formula intuition
slope = Delta y / Delta x

Python / system implementation idea
Estimate rolling slopes to detect changing exposure.

Failure-mode table

Mistake	Symptom	Fix
Invalid domain	NaN, inf, impossible values	Add validation before transform
Wrong timestamp	Backtest too good	Shift features; audit feature time
Overfit parameter	Only one setting works	Run sensitivity sweep

Where this fits in my research system
Risk attribution, factor sensitivity, optimization.

Confusions to avoid
Steep slope means high sensitivity, not automatically high predictive value.

Related terms
derivative, gradient, beta, delta

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FUNCTIONS

BEGINNER

Intercept / Baseline

One-sentence meaning

The intercept is the output when input equals zero.

Memory jogger

Starting level before x contributes.

Why it matters in quant research

In regression, intercept can approximate residual average return after exposures, but it is fragile.

Simple market / trading example

A strategy return model r = alpha + beta*mkt: alpha is baseline not explained by market.

Formula intuition

$y = b$ when $x = 0$

Python / system implementation idea

Center features to make intercept easier to interpret.

Quick recall check

Name the input, output, valid domain, likely quant use, and one production bug this concept could create if implemented carelessly.

Where this fits in my research system

Regression, factor attribution, alpha analysis.

Confusions to avoid

Intercept interpretation depends on whether x=0 is meaningful.

Related terms

alpha, constant, baseline

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FUNCTIONS

BEGINNER

Nonlinear Function

One-sentence meaning

A function where output does not change at a constant rate.

Memory jogger

The rule bends.

Why it matters in quant research

Markets often have thresholds, saturation, convex costs, nonlinear risk, and asymmetric reactions.

Simple market / trading example

Volatility impact may increase sharply when leverage rises past a risk limit.

Formula intuition

y = x^2, y = exp(x), y = log(x)

Python / system implementation idea

Compare nonlinear feature transforms with out-of-sample validation.

Inline workflow diagram

input → function → output → diagnostic → decision

Read the concept as one node in a larger pipeline. The key question is always whether its input is valid and timestamp-safe.

Where this fits in my research system

Feature transforms, payoff curves, risk constraints.

Confusions to avoid

Nonlinear does not automatically mean complex or predictive.

Related terms

curvature, convexity, interaction

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FUNCTIONS

BEGINNER

Quadratic Function

One-sentence meaning

A second-degree function with a parabolic shape.

Memory jogger

U-shape or upside-down U-shape.

Why it matters in quant research

Useful for convex penalties, squared error, variance, and simple curvature intuition.

Simple market / trading example

Mean squared error penalizes large forecast errors more than small errors.

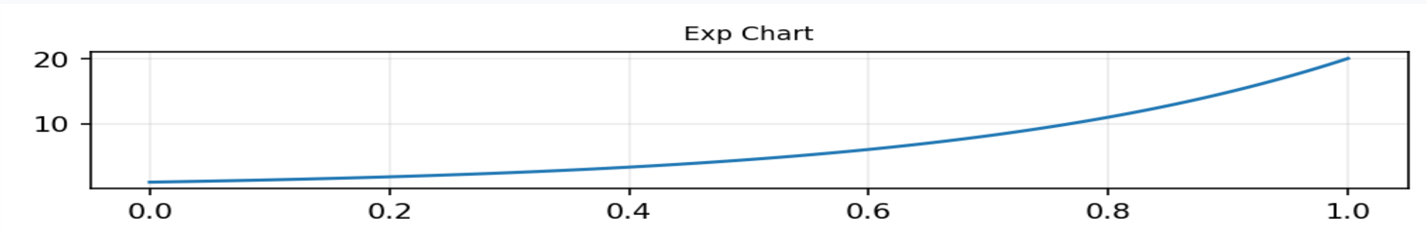
Formula intuition

y = ax^2 + bx + c

Python / system implementation idea

Use squared residual plots to identify outliers dominating loss.

Visual intuition



Synthetic chart for shape intuition; use it as a pattern detector, not as market evidence.

Where this fits in my research system

Loss functions, variance, convex risk penalties.

Confusions to avoid

Squaring removes sign; positive and negative errors both become positive.

Related terms

parabola, squared error, convex

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FUNCTIONS

BEGINNER

Exponential Function

One-sentence meaning

A function where change compounds multiplicatively.

Memory jogger

Growth feeds on itself.

Why it matters in quant research

Compounding, drawdowns, leverage growth, and continuous returns all use exponential intuition.

Simple market / trading example

Capital after log returns: equity_T = equity_0 * exp(sum r_log).

Formula intuition

y = a*b^x; y=e^x

Python / system implementation idea

Use log-space calculations to avoid numerical overflow.

Failure-mode table

Mistake	Symptom	Fix
Invalid domain	NaN, inf, impossible values	Add validation before transform
Wrong timestamp	Backtest too good	Shift features; audit feature time
Overfit parameter	Only one setting works	Run sensitivity sweep

Where this fits in my research system

Compounding, portfolio growth, risk of ruin.

Confusions to avoid

Exponential growth is unstable under small assumption errors.

Related terms

compound growth, e, log return

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FUNCTIONS

BEGINNER

Logarithmic Function

One-sentence meaning

A function that grows quickly at first then slowly; inverse of exponential.

Memory jogger

Compress big values; expand small differences.

Why it matters in quant research

Log transforms stabilize multiplicative data, convert ratios into additive returns, and reduce skew.

Simple market / trading example

log(P_t/P_{t-1}) gives log return.

Formula intuition

ln(a/b)=ln(a)-ln(b)

Python / system implementation idea

Validate positivity before log transforms.

Quick recall check

Name the input, output, valid domain, likely quant use, and one production bug this concept could create if implemented carelessly.

Where this fits in my research system

Return engineering, normalization, scale compression.

Confusions to avoid

Log requires positive input; zero or negative prices break it.

Related terms

natural log, log return, inverse

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FUNCTIONS

BEGINNER

Power Functions and Scaling

One-sentence meaning

Power functions raise input to a fixed exponent and control curvature.

Memory jogger

Exponent decides how aggressive the bend is.

Why it matters in quant research

Useful for volatility scaling, nonlinear penalties, and transforming skewed raw variables.

Simple market / trading example

Position size might scale as signal^0.5 to damp extreme signals.

Formula intuition

y=x^p

Python / system implementation idea

Log every transform parameter with experiment metadata.

Inline workflow diagram

input → function → output → diagnostic → decision

Read the concept as one node in a larger pipeline. The key question is always whether its input is valid and timestamp-safe.

Where this fits in my research system

Signal shaping, risk penalties, normalization.

Confusions to avoid

Changing scale can change rank, magnitude, and optimization behavior.

Related terms

square root, polynomial, exponent

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FUNCTIONS

BEGINNER

Piecewise Function

One-sentence meaning

A function defined by different rules over different input regions.

Memory jogger

Different rule depending on the zone.

Why it matters in quant research

Trading rules often use regimes, thresholds, stops, fees, and if/else logic.

Simple market / trading example

If volatility < 20%, use normal size; if volatility >= 20%, cut size in half.

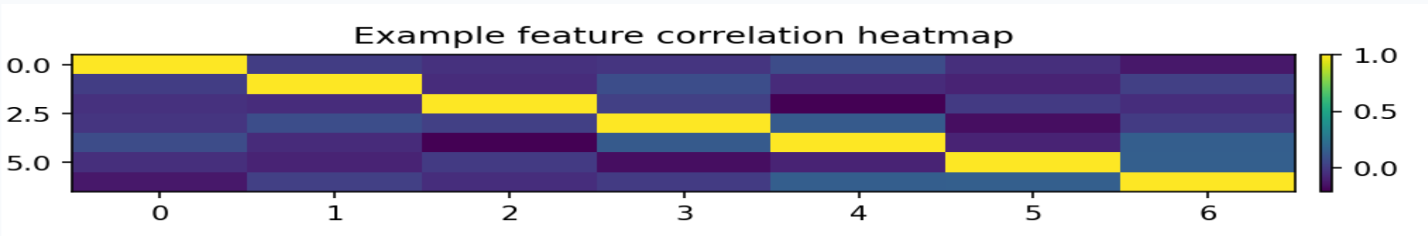
Formula intuition

f(x)=a if x=c

Python / system implementation idea

Unit-test every branch and boundary condition.

Visual intuition



Synthetic chart for shape intuition; use it as a pattern detector, not as market evidence.

Where this fits in my research system

Regime logic, execution rules, risk limits.

Confusions to avoid

Sharp boundaries can overfit and create unstable signals near cutoffs.

Related terms

threshold, branch, regime, indicator function

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FUNCTIONS

BEGINNER

Step Function

One-sentence meaning

A function that stays flat then jumps at thresholds.

Memory jogger

Flat shelves with jumps.

Why it matters in quant research

Useful for discrete signals, bins, quantiles, fee tiers, and rule triggers.

Simple market / trading example

Signal = 1 if momentum rank is top decile, else 0.

Formula intuition

I(x>c)

Python / system implementation idea

Track turnover introduced by threshold crossings.

Failure-mode table

Mistake	Symptom	Fix
Invalid domain	NaN, inf, impossible values	Add validation before transform
Wrong timestamp	Backtest too good	Shift features; audit feature time
Overfit parameter	Only one setting works	Run sensitivity sweep

Where this fits in my research system

Signal discretization, bucketing, alerts.

Confusions to avoid

Step functions lose magnitude information inside each bin.

Related terms

indicator, binning, threshold

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Indicator Function

One-sentence meaning

A function returning 1 when a condition is true and 0 otherwise.

Memory jogger

Boolean condition turned into a number.

Why it matters in quant research

Lets logical conditions enter formulas, filters, and models.

Simple market / trading example

I(volume_t > average_volume_20) flags unusually active days.

Formula intuition

1_A(x)=1 if x in A else 0

Python / system implementation idea

Use boolean masks in pandas and name conditions clearly.

Quick recall check

Name the input, output, valid domain, likely quant use, and one production bug this concept could create if implemented carelessly.

Where this fits in my research system

Feature flags, filters, regimes.

Confusions to avoid

Indicator variables are not probabilities unless calibrated that way.

Related terms

dummy variable, mask, boolean

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Absolute Value Function

One-sentence meaning

A function measuring distance from zero without sign.

Memory jogger

How far, not which direction.

Why it matters in quant research

Used for error magnitude, spread deviation, turnover, and transaction costs.

Simple market / trading example

Cost may depend on abs(position_t - position_{t-1}).

Formula intuition

$|x|$

Python / system implementation idea

Calculate turnover as abs(delta_position).sum().

Inline workflow diagram

input → function → output → diagnostic → decision

Read the concept as one node in a larger pipeline. The key question is always whether its input is valid and timestamp-safe.

Where this fits in my research system

Loss metrics, risk, costs.

Confusions to avoid

Absolute value hides direction; pair it with sign when direction matters.

Related terms

magnitude, distance, L1 loss

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Sign Function

One-sentence meaning
A function returning direction: negative, zero, or positive.

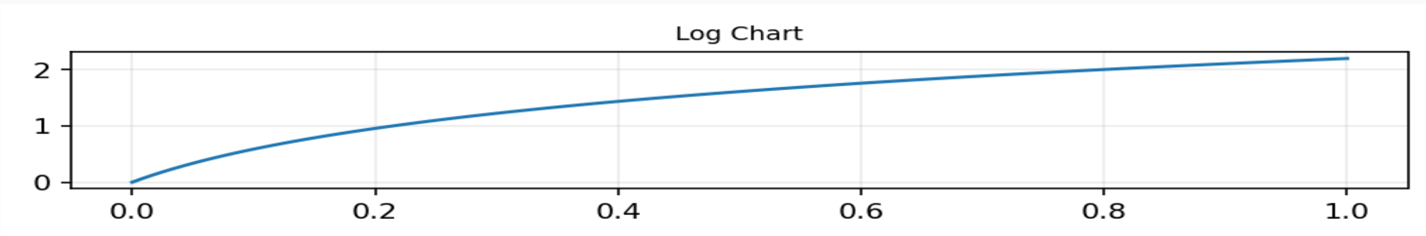
Memory jogger
Which side of zero are you on?

Why it matters in quant research
Converts continuous estimates into directional trading stances.

Simple market / trading example
sign(predicted_return) maps forecasts into long, flat, or short direction.

Formula intuition
sign(x) in {-1,0,1}

Python / system implementation idea
Keep raw forecast and signed signal for diagnostics.

Visual intuition


Synthetic chart for shape intuition; use it as a pattern detector, not as market evidence.

Where this fits in my research system
Signal generation, classification, directional bets.

Confusions to avoid
Sign discards confidence and magnitude.

Related terms
direction, threshold, long/short

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Backtest role	Whether the function creates features, targets, signals, positions, costs, or diagnostics.
Robustness	Neighboring parameters and adjacent regimes do not completely reverse the conclusion.

Mini recall prompt
Explain this concept in one sentence, name one formula or code representation, and describe one way it can create a false backtest if misaligned.

System-design connection
Treat this as a node with an explicit input schema, output schema, parameters, version, and timestamp rule. That is what makes research reproducible.

FUNCTIONS

BEGINNER

Min, Max, Clip

One-sentence meaning

Functions that cap values below, above, or both.

Memory jogger

Guardrails for extreme values.

Why it matters in quant research

Controls leverage, outliers, risk limits, and numerical instability.

Simple market / trading example

Clip signal between -1 and +1 before position sizing.

Formula intuition

clip(x,a,b)=min(max(x,a),b)

Python / system implementation idea

Apply clipping after recording raw values for auditability.

Failure-mode table

Mistake	Symptom	Fix
Invalid domain	NaN, inf, impossible values	Add validation before transform
Wrong timestamp	Backtest too good	Shift features; audit feature time
Overfit parameter	Only one setting works	Run sensitivity sweep

Where this fits in my research system

Risk controls, preprocessing, execution caps.

Confusions to avoid

Clipping can hide real stress events if overused.

Related terms

winsorize, cap, floor, constraint

Validation / implementation checklist

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FUNCTIONS

BEGINNER

Inverse Function

One-sentence meaning
An inverse reverses a mapping where possible.

Memory jogger
Undo button for a function.

Why it matters in quant research
Needed for transforming data back to original scale and interpreting models trained on transformed values.

Simple market / trading example
If y=ln(P), then P=exp(y).

Formula intuition
f^{-1}(f(x))=x

Python / system implementation idea
Store scaler objects so predictions can be inverse-transformed.

Quick recall check
Name the input, output, valid domain, likely quant use, and one production bug this concept could create if implemented carelessly.

Where this fits in my research system
Normalization inversion, price reconstruction, calibration.

Confusions to avoid
Only one-to-one functions have true inverses over their domain.

Related terms
one-to-one, invertible, exp/log

Validation / implementation checklist

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Mini recall prompt
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FUNCTIONS

BEGINNER

Composition of Functions

One-sentence meaning
Composition feeds one function's output into another function.

Memory jogger
Output of one machine becomes input to the next.

Why it matters in quant research
Research pipelines are composed functions from raw data to signals to trades.

Simple market / trading example
position = size(clip(rank(momentum(prices)))).

Formula intuition
$(g \circ f)(x)=g(f(x))$

Python / system implementation idea
Represent pipelines as ordered transforms with versioned configs.

Inline workflow diagram

input → function → output → diagnostic → decision

Read the concept as one node in a larger pipeline. The key question is always whether its input is valid and timestamp-safe.

Where this fits in my research system
Pipeline design, feature engineering, model stack.

Confusions to avoid
Order matters: normalize(rank(x)) is not rank(normalize(x)).

Related terms
pipeline, chain, nested function

Validation / implementation checklist

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FUNCTIONS

BEGINNER

Transformation

One-sentence meaning

A transformation changes representation while preserving or emphasizing useful structure.

Memory jogger

Same data, new lens.

Why it matters in quant research

Transforms make raw price/volume data usable for modeling and comparison.

Simple market / trading example

Raw prices -> returns -> z-scores -> ranks.

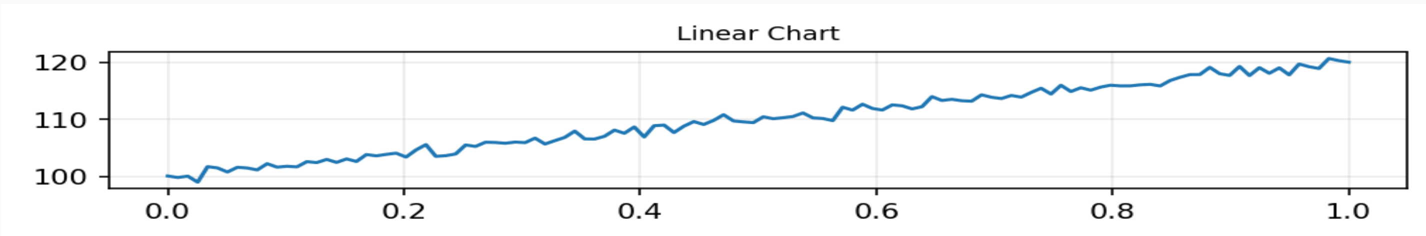
Formula intuition

$x \to T(x)$

Python / system implementation idea

Fit transforms on train only; apply to validation/test.

Visual intuition



Synthetic chart for shape intuition; use it as a pattern detector, not as market evidence.

Where this fits in my research system

Feature engineering, normalization, stationarity.

Confusions to avoid

Transforming can create leakage if future data enters the calculation.

Related terms

mapping, preprocessing, feature transform

Validation / implementation checklist

Check	What to verify
Input contract	Correct units, allowed domain, no missing values, timestamp-safe availability.
Output contract	Expected shape, range, sign convention, and no impossible values.
Backtest role	Whether the function creates features, targets, signals, positions, costs, or diagnostics.
Robustness	Neighboring parameters and adjacent regimes do not completely reverse the conclusion.

Mini recall prompt

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System-design connection

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FUNCTIONS

Translation / Shift

BEGINNER

One-sentence meaning

Adding or subtracting a constant moves a graph up/down or left/right.

Memory jogger

Same shape, moved location.

Why it matters in quant research

Mean-centering and lagging are practical shifts in quant systems.

Simple market / trading example

Subtract rolling mean from price spread to create deviation from normal.

Formula intuition

$y=f(x-c)+d$

Python / system implementation idea

Use explicit shift(1) when features must be known before trading.

Failure-mode table

Mistake	Symptom	Fix
Invalid domain	NaN, inf, impossible values	Add validation before transform
Wrong timestamp	Backtest too good	Shift features; audit feature time
Overfit parameter	Only one setting works	Run sensitivity sweep

Where this fits in my research system

Centering, lag features, detrending.

Confusions to avoid

A time shift can accidentally use future values if direction is wrong.

Related terms

offset, centering, lag

Validation / implementation checklist

Check	What to verify
Input contract	Correct units, allowed domain, no missing values, timestamp-safe availability.
Output contract	Expected shape, range, sign convention, and no impossible values.
Backtest role	Whether the function creates features, targets, signals, positions, costs, or diagnostics.
Robustness	Neighboring parameters and adjacent regimes do not completely reverse the conclusion.

Mini recall prompt

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System-design connection

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FUNCTIONS

BEGINNER

Scaling / Stretching

One-sentence meaning

Multiplying input or output stretches or compresses a function.

Memory jogger

Same shape, different units or intensity.

Why it matters in quant research

Scaling affects model coefficients, optimization, leverage, and feature comparability.

Simple market / trading example

Standardize returns by volatility so assets with different risk can be compared.

Formula intuition

y=a*f(bx)

Python / system implementation idea

Track units: dollars, returns, bps, z-scores.

Quick recall check

Name the input, output, valid domain, likely quant use, and one production bug this concept could create if implemented carelessly.

Where this fits in my research system

Normalization, volatility targeting, risk parity.

Confusions to avoid

Scale changes magnitude but not always ranking.

Related terms

standardization, leverage, units

Validation / implementation checklist

Check	What to verify
Input contract	Correct units, allowed domain, no missing values, timestamp-safe availability.
Output contract	Expected shape, range, sign convention, and no impossible values.
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Mini recall prompt

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System-design connection

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FUNCTIONS

BEGINNER

Reflection

One-sentence meaning
A reflection flips a graph across an axis.

Memory jogger
Mirror image.

Why it matters in quant research
Useful for short-side logic, inverse relationships, and sign conventions.

Simple market / trading example
A short position has PnL roughly reflected against the long position before costs.

Formula intuition
y=-f(x) or y=f(-x)

Python / system implementation idea
Backtest long and short legs separately.

Inline workflow diagram

input → function → output → diagnostic → decision

Read the concept as one node in a larger pipeline. The key question is always whether its input is valid and timestamp-safe.

Where this fits in my research system
Long/short design, payoff intuition.

Confusions to avoid
Flipping signal direction is not the same as proving alpha on the short side.

Related terms
negative sign, inverse exposure

Validation / implementation checklist

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Mini recall prompt
Explain this concept in one sentence, name one formula or code representation, and describe one way it can create a false backtest if misaligned.

System-design connection
Treat this as a node with an explicit input schema, output schema, parameters, version, and timestamp rule. That is what makes research reproducible.

FUNCTIONS

BEGINNER

Normalization

One-sentence meaning
Normalization rescales values so they are comparable across assets or periods.

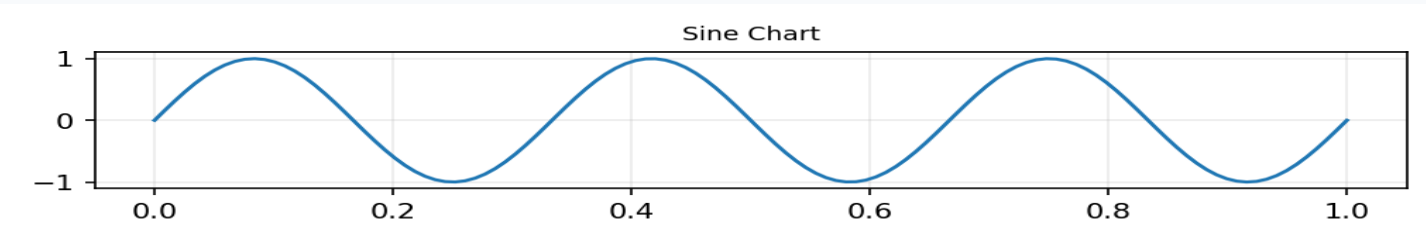
Memory jogger
Put different units onto a common scale.

Why it matters in quant research
Prevents high-priced or volatile assets from dominating models just because of scale.

Simple market / trading example
Convert returns to z-scores using rolling mean and rolling standard deviation.

Formula intuition
z=(x-mu)/sigma

Python / system implementation idea
Use rolling or train-fitted scalers, not full-sample scalers.

Visual intuition


Synthetic chart for shape intuition; use it as a pattern detector, not as market evidence.

Where this fits in my research system
Preprocessing, cross-sectional ranking, ML features.

Confusions to avoid
Normalize using only information available at that time.

Related terms
standardize, rescale, z-score

Validation / implementation checklist

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Input contract	Correct units, allowed domain, no missing values, timestamp-safe availability.
Output contract	Expected shape, range, sign convention, and no impossible values.
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System-design connection
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FUNCTIONS

BEGINNER

Z-Score Function

One-sentence meaning

A z-score measures how many standard deviations a value is from its mean.

Memory jogger

Distance from normal in volatility units.

Why it matters in quant research

Common for mean reversion, anomaly detection, and feature normalization.

Simple market / trading example

Spread z-score = (spread - rolling_mean) / rolling_std.

Formula intuition

z_t=(x_t-mu_t)/sigma_t

Python / system implementation idea

Compute rolling z-scores with min periods and no future data.

Failure-mode table

Mistake	Symptom	Fix
Invalid domain	NaN, inf, impossible values	Add validation before transform
Wrong timestamp	Backtest too good	Shift features; audit feature time
Overfit parameter	Only one setting works	Run sensitivity sweep

Where this fits in my research system

Mean reversion, outlier flags, cross-asset comparison.

Confusions to avoid

A z-score assumes mean/std are meaningful in the current regime.

Related terms

standard deviation, anomaly, stationarity

Validation / implementation checklist

Check	What to verify
Input contract	Correct units, allowed domain, no missing values, timestamp-safe availability.
Output contract	Expected shape, range, sign convention, and no impossible values.
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FUNCTIONS

BEGINNER

Rank Transform

One-sentence meaning

A rank transform replaces values with their order position.

Memory jogger

Who is higher, not by how much.

Why it matters in quant research

Robust to outliers and common in cross-sectional equity signals.

Simple market / trading example

Rank stocks by 12-month momentum and buy the top decile.

Formula intuition

rank(x_i)

Python / system implementation idea

Use groupby(date).rank(pct=True) for cross-sectional ranks.

Quick recall check

Name the input, output, valid domain, likely quant use, and one production bug this concept could create if implemented carelessly.

Where this fits in my research system

Cross-sectional signals, factor portfolios.

Confusions to avoid

Ranks discard magnitude; rank 1 vs 2 may be tiny or huge.

Related terms

percentile, quantile, order statistic

Validation / implementation checklist

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System-design connection

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FUNCTIONS

BEGINNER

Percent Change Function

One-sentence meaning

Percent change measures relative change from one value to the next.

Memory jogger

How much did it move compared with where it started?

Why it matters in quant research

Turns prices into comparable return-like values across assets.

Simple market / trading example

A stock from 100 to 105 has a 5% simple return.

Formula intuition

$r_t=P_t/P_{t-1}-1$

Python / system implementation idea

Use pct_change() and align labels carefully.

Inline workflow diagram

input → function → output → diagnostic → decision

Read the concept as one node in a larger pipeline. The key question is always whether its input is valid and timestamp-safe.

Where this fits in my research system

Return calculation, feature construction.

Confusions to avoid

Percent change compounds multiplicatively over multiple periods.

Related terms

simple return, arithmetic return

Validation / implementation checklist

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FUNCTIONS

BEGINNER

Log Return Function

One-sentence meaning
Log return is the logarithm of the price ratio and adds through time.

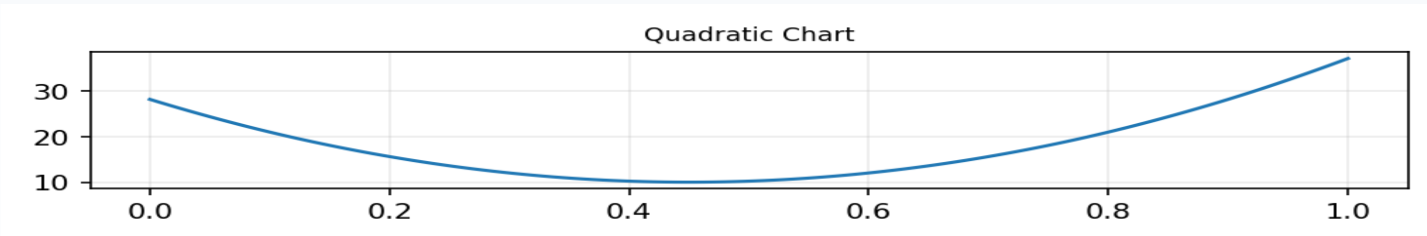
Memory jogger
Continuous compounding return.

Why it matters in quant research
Useful for aggregation, modeling, and mathematically cleaner return sums.

Simple market / trading example
Two log returns can be summed to get total log return over two days.

Formula intuition
r_t=ln(P_t/P_{t-1})

Python / system implementation idea
Use np.log(price).diff() and handle missing prices.

Visual intuition


Synthetic chart for shape intuition; use it as a pattern detector, not as market evidence.

Where this fits in my research system
Return engineering, compounding, time aggregation.

Confusions to avoid
Log returns approximate simple returns only for small moves.

Related terms
continuous return, ln, compounding

Validation / implementation checklist

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FUNCTIONS

BEGINNER

Cumulative Return Function

One-sentence meaning

Cumulative return aggregates period returns into total growth.

Memory jogger

Turn many small moves into one total move.

Why it matters in quant research

Measures strategy equity curve, drawdowns, and realized growth.

Simple market / trading example

If daily returns are 1%, -1%, 2%, cumulative simple return is (1.01*0.99*1.02)-1.

Formula intuition

R_cum = prod(1+r_t)-1

Python / system implementation idea

Compute equity=(1+returns).cumprod().

Failure-mode table

Mistake	Symptom	Fix
Invalid domain	NaN, inf, impossible values	Add validation before transform
Wrong timestamp	Backtest too good	Shift features; audit feature time
Overfit parameter	Only one setting works	Run sensitivity sweep

Where this fits in my research system

Backtesting, performance reporting.

Confusions to avoid

Summing simple returns is wrong for cumulative performance.

Related terms

equity curve, compounding, total return

Validation / implementation checklist

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FUNCTIONS

BEGINNER

Moving Average Function

One-sentence meaning

A moving average maps a time series to a smoothed version using a rolling window.

Memory jogger

Rolling local average.

Why it matters in quant research

Used for smoothing, trend filters, baselines, and mean-reversion deviations.

Simple market / trading example

20-day moving average of price is a local trend estimate.

Formula intuition

MA_t=(1/n) sum_{i=0}^{n-1} x_{t-i}

Python / system implementation idea

Use rolling(window).mean().shift(1) for trade-safe features.

Quick recall check

Name the input, output, valid domain, likely quant use, and one production bug this concept could create if implemented carelessly.

Where this fits in my research system

Trend filters, features, noise reduction.

Confusions to avoid

Moving averages lag; they do not predict by themselves.

Related terms

rolling mean, smoothing, lag

Validation / implementation checklist

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System-design connection

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FUNCTIONS

BEGINNER

Rolling Window Function

One-sentence meaning

A rolling function applies a calculation to the most recent n observations.

Memory jogger

A moving local calculator.

Why it matters in quant research

Markets change; rolling windows estimate local state instead of full-history averages.

Simple market / trading example

Rolling 60-day volatility estimates current risk.

Formula intuition

$F_t = h(x_{t-n+1},...,x_t)$

Python / system implementation idea

Log window sizes and test robustness across neighboring values.

Inline workflow diagram

input → function → output → diagnostic → decision

Read the concept as one node in a larger pipeline. The key question is always whether its input is valid and timestamp-safe.

Where this fits in my research system

Feature engineering, risk models, validation.

Confusions to avoid

Window length is a parameter that can overfit.

Related terms

window, lookback, trailing sample

Validation / implementation checklist

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Output contract	Expected shape, range, sign convention, and no impossible values.
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Mini recall prompt

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System-design connection

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FUNCTIONS

BEGINNER

Lag Function

One-sentence meaning

A lag shifts data backward so past values line up with current decisions.

Memory jogger

Yesterday's value placed next to today.

Why it matters in quant research

Prevents lookahead leakage and builds autoregressive features.

Simple market / trading example

Use return_{t-1} to predict return_t.

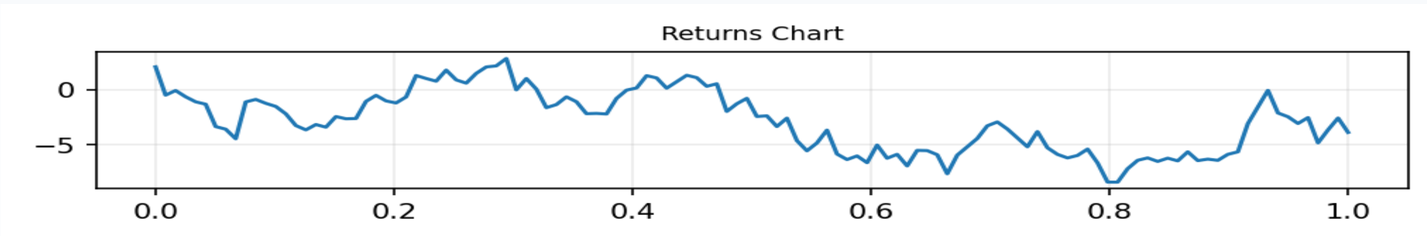
Formula intuition

lag_k(x_t)=x_{t-k}

Python / system implementation idea

Require feature timestamp <= decision timestamp.

Visual intuition



Synthetic chart for shape intuition; use it as a pattern detector, not as market evidence.

Where this fits in my research system

Feature alignment, time-series modeling.

Confusions to avoid

Wrong lag direction is one of the fastest ways to fake performance.

Related terms

shift, delay, autocorrelation

Validation / implementation checklist

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System-design connection

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FUNCTIONS

BEGINNER

Lead Function

One-sentence meaning

A lead shifts future values backward for target construction, not features.

Memory jogger

Tomorrow's outcome labeled beside today's features.

Why it matters in quant research

Used to create prediction labels while keeping features strictly historical.

Simple market / trading example

target_t = return_{t+1}.

Formula intuition

lead_k(x_t)=x_{t+k}

Python / system implementation idea

Keep target columns named target_* and exclude them from X.

Failure-mode table

Mistake	Symptom	Fix
Invalid domain	NaN, inf, impossible values	Add validation before transform
Wrong timestamp	Backtest too good	Shift features; audit feature time
Overfit parameter	Only one setting works	Run sensitivity sweep

Where this fits in my research system

Supervised learning target generation.

Confusions to avoid

Lead values must never be used as input features.

Related terms

label, forecast horizon, future return

Validation / implementation checklist

Check	What to verify
Input contract	Correct units, allowed domain, no missing values, timestamp-safe availability.
Output contract	Expected shape, range, sign convention, and no impossible values.
Backtest role	Whether the function creates features, targets, signals, positions, costs, or diagnostics.
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Mini recall prompt

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System-design connection

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FUNCTIONS

BEGINNER

Difference Function

One-sentence meaning
A difference subtracts consecutive values to measure change in level.

Memory jogger
Today minus yesterday.

Why it matters in quant research
Can remove trend and make some series easier to model.

Simple market / trading example
Price difference $P_t - P_{t-1}$; spread difference for stationarity tests.

Formula intuition
$\Delta x_t = x_t - x_{t-1}$

Python / system implementation idea
Use diff() for levels; pct_change/log diff for returns.

Quick recall check
Name the input, output, valid domain, likely quant use, and one production bug this concept could create if implemented carelessly.

Where this fits in my research system
Stationarity, deltas, trend removal.

Confusions to avoid
Price differences are not scale-comparable across assets.

Related terms
delta, first difference, differencing

Validation / implementation checklist

Check	What to verify
Input contract	Correct units, allowed domain, no missing values, timestamp-safe availability.
Output contract	Expected shape, range, sign convention, and no impossible values.
Backtest role	Whether the function creates features, targets, signals, positions, costs, or diagnostics.
Robustness	Neighboring parameters and adjacent regimes do not completely reverse the conclusion.

Mini recall prompt
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System-design connection
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FUNCTIONS

BEGINNER

Smoothing Function

One-sentence meaning

Smoothing reduces high-frequency noise to reveal lower-frequency structure.

Memory jogger

Quiet the noise without pretending it disappeared.

Why it matters in quant research

Useful for noisy indicators, volatility estimates, and visualization; dangerous when it adds lag.

Simple market / trading example

Exponential moving average reacts faster than simple moving average.

Formula intuition

$EMA_t = alpha*x_t + (1-alpha)*EMA_{t-1}$

Python / system implementation idea

Compare raw and smoothed signal turnover.

Inline workflow diagram

input → function → output → diagnostic → decision

Read the concept as one node in a larger pipeline. The key question is always whether its input is valid and timestamp-safe.

Where this fits in my research system

Indicators, denoising, trend extraction.

Confusions to avoid

Smoothing can make signals look better visually while delaying action.

Related terms

EMA, filter, rolling mean

Validation / implementation checklist

Check	What to verify
Input contract	Correct units, allowed domain, no missing values, timestamp-safe availability.
Output contract	Expected shape, range, sign convention, and no impossible values.
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Robustness	Neighboring parameters and adjacent regimes do not completely reverse the conclusion.

Mini recall prompt

Explain this concept in one sentence, name one formula or code representation, and describe one way it can create a false backtest if misaligned.

System-design connection

Treat this as a node with an explicit input schema, output schema, parameters, version, and timestamp rule. That is what makes research reproducible.

FUNCTIONS

BEGINNER

Loss Function

One-sentence meaning

A loss function scores how wrong a prediction or decision is.

Memory jogger

Penalty meter for mistakes.

Why it matters in quant research

Defines what a model optimizes; wrong loss means wrong behavior.

Simple market / trading example

MSE punishes large return forecast errors; asymmetric loss can punish downside more.

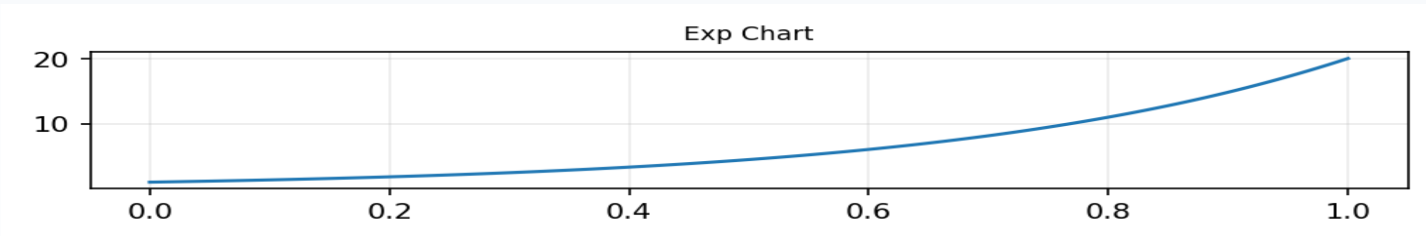
Formula intuition

MSE = mean((y-yhat)^2)

Python / system implementation idea

Evaluate loss by time split and market regime.

Visual intuition



Synthetic chart for shape intuition; use it as a pattern detector, not as market evidence.

Where this fits in my research system

Model training, optimization, validation.

Confusions to avoid

Low training loss does not imply tradable alpha.

Related terms

objective, error, cost function

Validation / implementation checklist

Check	What to verify
Input contract	Correct units, allowed domain, no missing values, timestamp-safe availability.
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FUNCTIONS

BEGINNER

Utility Function

One-sentence meaning

A utility function scores preferences, not just raw profit.

Memory jogger

What you actually care about encoded as a score.

Why it matters in quant research

Trading systems should optimize risk-adjusted, cost-aware objectives, not only return.

Simple market / trading example

Utility = return - lambda * variance - costs.

Formula intuition

U = E[R] - lambda*Var(R)

Python / system implementation idea

Separate research metric, training objective, and production risk limits.

Failure-mode table

Mistake	Symptom	Fix
Invalid domain	NaN, inf, impossible values	Add validation before transform
Wrong timestamp	Backtest too good	Shift features; audit feature time
Overfit parameter	Only one setting works	Run sensitivity sweep

Where this fits in my research system

Portfolio optimization, objective design.

Confusions to avoid

Utility is subjective; parameter choices encode preferences.

Related terms

risk aversion, objective, reward

Validation / implementation checklist

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FUNCTIONS

BEGINNER

Constraint Function

One-sentence meaning

A constraint restricts allowed solutions or actions.

Memory jogger

The rules of the sandbox.

Why it matters in quant research

Keeps optimizers and strategies inside risk, leverage, liquidity, and operational limits.

Simple market / trading example

Sum of absolute weights <= 1.5 controls gross leverage.

Formula intuition

$g(x) \le 0$, $h(x)=0$

Python / system implementation idea

Validate proposed orders against constraints before sending.

Quick recall check

Name the input, output, valid domain, likely quant use, and one production bug this concept could create if implemented carelessly.

Where this fits in my research system

Portfolio construction, execution, risk management.

Confusions to avoid

Unmodeled constraints appear later as production failures.

Related terms

bounds, feasible set, optimization

Validation / implementation checklist

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FUNCTIONS

BEGINNER

Objective Function

One-sentence meaning

An objective function is what an algorithm tries to maximize or minimize.

Memory jogger

The score the machine chases.

Why it matters in quant research

Objective choice determines strategy behavior and failure modes.

Simple market / trading example

Maximize expected return minus transaction costs and risk penalty.

Formula intuition

argmax_theta J(theta)

Python / system implementation idea

Use walk-forward validation and parameter stability checks.

Inline workflow diagram

input → function → output → diagnostic → decision

Read the concept as one node in a larger pipeline. The key question is always whether its input is valid and timestamp-safe.

Where this fits in my research system

Optimization, model selection, parameter search.

Confusions to avoid

Optimizing noisy backtest Sharpe can select overfit parameters.

Related terms

loss, utility, fitness, reward

Validation / implementation checklist

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FUNCTIONS

BEGINNER

Parameter vs Variable

One-sentence meaning

Variables change as inputs; parameters are fixed settings controlling the function.

Memory jogger

Input is fed in; parameter tunes the machine.

Why it matters in quant research

Distinguishing them keeps experiments reproducible and prevents accidental tuning on test data.

Simple market / trading example

Lookback=20 is a parameter; today's return is a variable input.

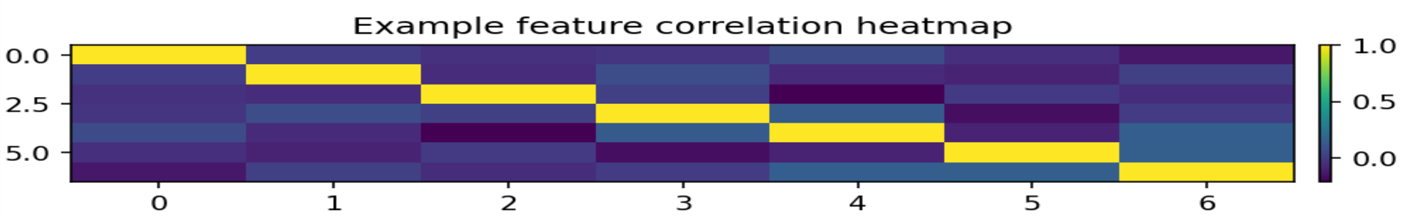
Formula intuition

f(x; theta)

Python / system implementation idea

Version every parameter set with the backtest result.

Visual intuition



Synthetic chart for shape intuition; use it as a pattern detector, not as market evidence.

Where this fits in my research system

Experiment design, hyperparameter search.

Confusions to avoid

A learned coefficient can become a parameter after training.

Related terms

hyperparameter, coefficient, config

Validation / implementation checklist

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FUNCTIONS

BEGINNER

Sensitivity Analysis

One-sentence meaning

Sensitivity analysis studies how outputs change when inputs or parameters change.

Memory jogger

Shake the inputs and see what breaks.

Why it matters in quant research

Reveals fragile features, overfit thresholds, unstable parameters, and hidden risk.

Simple market / trading example

Test Sharpe for moving-average windows 15, 20, 25 rather than only 20.

Formula intuition

S = Delta output / Delta input

Python / system implementation idea

Run grid sweeps and heatmaps for core parameters.

Failure-mode table

Mistake	Symptom	Fix
Invalid domain	NaN, inf, impossible values	Add validation before transform
Wrong timestamp	Backtest too good	Shift features; audit feature time
Overfit parameter	Only one setting works	Run sensitivity sweep

Where this fits in my research system

Robustness testing, diagnostics.

Confusions to avoid

A good result at one exact parameter is weak evidence.

Related terms

gradient, perturbation, parameter sweep

Validation / implementation checklist

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FUNCTIONS

BEGINNER

Continuity

One-sentence meaning

A continuous function has no jumps; small input changes create small output changes.

Memory jogger

No sudden teleporting in the graph.

Why it matters in quant research

Continuous transforms are often more stable than hard thresholds in noisy markets.

Simple market / trading example

Position size that changes smoothly with z-score reduces churn near boundaries.

Formula intuition

lim_{x->a} f(x)=f(a)

Python / system implementation idea

Prefer smooth sizing functions when threshold turnover is high.

Quick recall check

Name the input, output, valid domain, likely quant use, and one production bug this concept could create if implemented carelessly.

Where this fits in my research system

Signal shaping, risk scaling.

Confusions to avoid

Continuous does not mean predictable.

Related terms

smoothness, jump, discontinuity

Validation / implementation checklist

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FUNCTIONS

BEGINNER

Monotonic Function

One-sentence meaning

A monotonic function moves in one direction: never decreasing or never increasing.

Memory jogger

More input never reverses the output direction.

Why it matters in quant research

Useful when a belief is ordinal, such as higher risk should not increase position size.

Simple market / trading example

Position size decreases monotonically as volatility increases.

Formula intuition

x1 f(x1)<=f(x2)

Python / system implementation idea

Check monotonic calibration curves for model outputs.

Inline workflow diagram

input → function → output → diagnostic → decision

Read the concept as one node in a larger pipeline. The key question is always whether its input is valid and timestamp-safe.

Where this fits in my research system

Risk controls, rankings, calibration.

Confusions to avoid

Monotonic transforms can still distort magnitudes.

Related terms

increasing, decreasing, order-preserving

Validation / implementation checklist

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FUNCTIONS

BEGINNER

Bounded Function

One-sentence meaning
A bounded function has outputs within a fixed range.

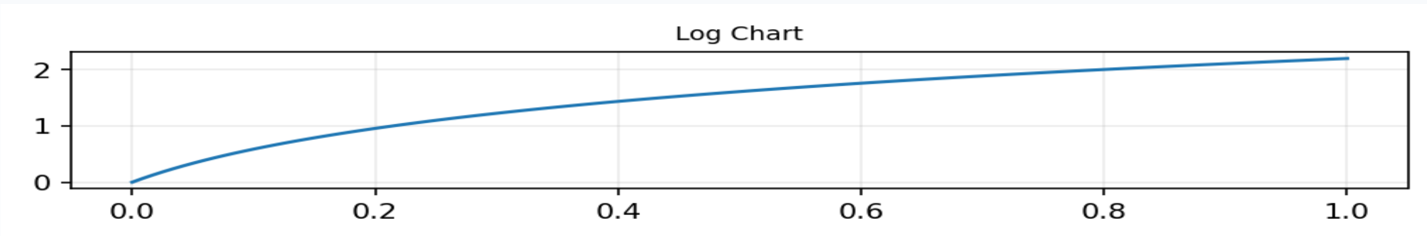
Memory jogger
Output lives inside guardrails.

Why it matters in quant research
Important for position sizing, probabilities, normalized signals, and risk caps.

Simple market / trading example
tanh(signal) maps any raw score into (-1,1).

Formula intuition
a <= f(x) <= b

Python / system implementation idea
Use bounded transforms before converting scores to leverage.

Visual intuition


Synthetic chart for shape intuition; use it as a pattern detector, not as market evidence.

Where this fits in my research system
Signal scaling, probability models, exposure caps.

Confusions to avoid
Bounded output can saturate and hide extreme input differences.

Related terms
clip, sigmoid, tanh, limits

Validation / implementation checklist

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FUNCTIONS

BEGINNER

Sigmoid and Tanh Functions

One-sentence meaning

Sigmoid maps to 0-1; tanh maps to -1 to 1 with smooth saturation.

Memory jogger

Soft threshold with capped output.

Why it matters in quant research

Useful for probabilities, smooth position sizing, and reducing outlier impact.

Simple market / trading example

A raw alpha score becomes position = tanh(score).

Formula intuition

sigmoid(x)=1/(1+e^-x); tanh(x)

Python / system implementation idea

Monitor distribution before and after saturation.

Failure-mode table

Mistake	Symptom	Fix
Invalid domain	NaN, inf, impossible values	Add validation before transform
Wrong timestamp	Backtest too good	Shift features; audit feature time
Overfit parameter	Only one setting works	Run sensitivity sweep

Where this fits in my research system

ML models, signal shaping, exposure control.

Confusions to avoid

Saturation can make very different scores produce nearly identical positions.

Related terms

logistic, saturation, activation

Validation / implementation checklist

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FUNCTIONS

BEGINNER

Function Pipeline in Quant Research

One-sentence meaning

A quant pipeline is a composed function from raw data to decisions and evaluation.

Memory jogger

Raw data goes through machines until it becomes a tested strategy.

Why it matters in quant research

Thinking in functions makes systems auditable, testable, modular, and less magical.

Simple market / trading example

prices -> returns -> features -> model -> signal -> portfolio -> backtest.

Formula intuition

$decision_t = F(data_{<=t}, params)$

Python / system implementation idea

Implement pipeline nodes with timestamps, configs, and immutable outputs.

Quick recall check

Name the input, output, valid domain, likely quant use, and one production bug this concept could create if implemented carelessly.

Where this fits in my research system

End-to-end research architecture.

Confusions to avoid

A pipeline can be mathematically valid and still leak future data.

Related terms

DAG, pipeline, transform graph

Validation / implementation checklist

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FUNCTIONS

BEGINNER

Graph Diagnostics for Signals

One-sentence meaning

Graph diagnostics plot functions, features, returns, and outputs to catch impossible behavior.

Memory jogger

Graphs are smoke detectors.

Why it matters in quant research

Many quant bugs are visible before they are statistically obvious.

Simple market / trading example

Plot signal against future returns, turnover, volatility regimes, and time.

Formula intuition

visual diagnostic = sampled view of f/data

Python / system implementation idea

Create a standard chart pack for every experiment.

Inline workflow diagram

input → function → output → diagnostic → decision

Read the concept as one node in a larger pipeline. The key question is always whether its input is valid and timestamp-safe.

Where this fits in my research system

Research QA, model inspection.

Confusions to avoid

Pretty charts can deceive if axes, samples, or survivorship are wrong.

Related terms

scatterplot, residuals, equity curve

Validation / implementation checklist

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FUNCTIONS

BEGINNER

Review: Functions to Research System

One-sentence meaning
Functions become the grammar of a quant research system.

Memory jogger
Everything is a mapping: data in, decision out.

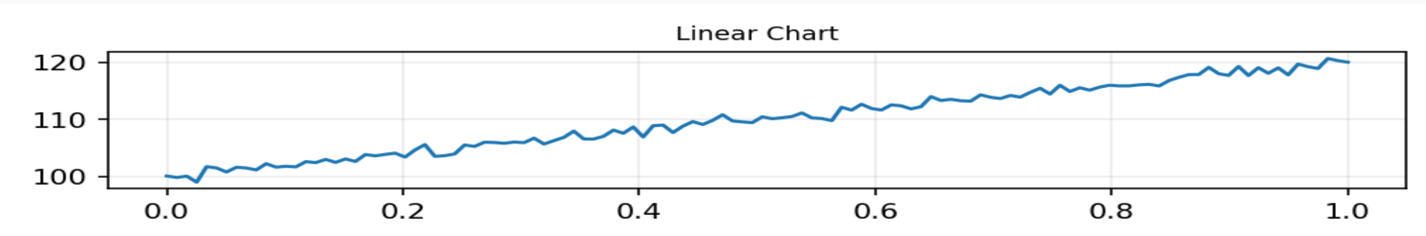
Why it matters in quant research
This review ties beginner function ideas to practical system design.

Simple market / trading example
A backtest is a function from historical data and parameters to performance diagnostics.

Formula intuition
backtest_result = B(data, strategy, costs, constraints)

Python / system implementation idea
Build reusable modules: validate -> transform -> model -> signal -> evaluate.

Visual intuition



Synthetic chart for shape intuition; use it as a pattern detector, not as market evidence.

Where this fits in my research system
Concept consolidation.

Confusions to avoid
Do not confuse clean notation with real-world validity.

Related terms
pipeline, objective, transform, validation

Validation / implementation checklist

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REVIEW

BEGINNER

Final Review and Index

Core synthesis

Functions are the control language of quant research: they define transformations, forecasts, objectives, constraints, diagnostics, and evaluation. A clean research system makes these mappings explicit, versioned, testable, and timestamp-safe.

Research pipeline map

validate data → engineer features → fit model → generate signal → size portfolio → backtest → diagnose

Concept family	Terms to recall	System question
Basic mappings	function, domain, range, notation	What is the input and output?
Graph behavior	linear, nonlinear, slope, intercept	How does output react to input?
Transforms	log, z-score, rank, clip, normalize	Does this make data more comparable or safer?
Time-series functions	lag, lead, rolling, moving average	Is the timestamp alignment valid?
Optimization	loss, utility, objective, constraint	What behavior is the system chasing?
Robustness	sensitivity, continuity, boundedness	Does the result survive perturbation?

Recall questions

1. Why is domain validation a trading-system issue, not just a math issue? 2. Why can a threshold rule create turnover? 3. Why should transformations be fitted on training data only? 4. How are objective functions related to overfitting? 5. Why is a backtest itself a function?

REVIEW

BEGINNER

Formula Summary Sheet

Formula	Meaning	Quant use
y = f(x)	Output is a function of input	General mapping
y = mx + b	Linear function	Factor exposure baseline
slope = Delta y / Delta x	Rate of change	Sensitivity / beta intuition
r_t = P_t/P_{t-1}-1	Simple return	Percent change
r_t = ln(P_t/P_{t-1})	Log return	Additive time aggregation
R = prod(1+r_t)-1	Cumulative return	Equity curve growth
z=(x-mu)/sigma	Z-score	Normalization / anomaly detection
MA_t=(1/n)sum x	Moving average	Smoothing / trend filter
EMA_t=alpha x_t +(1-alpha)EMA_{t-1}	Exponential smoothing	Adaptive smoothing
argmax_theta J(theta)	Optimization	Parameter selection

Notation table

Symbol	Plain-English meaning
x	Input variable or feature
y	Output value or target
f, g, h	Named functions or transforms
theta	Parameter set / learned weights / hyperparameters
P_t	Price at time t
r_t	Return at time t
mu, sigma	Mean and standard deviation
w	Portfolio weights
J(theta)	Objective score being optimized

Worked calculation

Simple return from 100 to 105: r = 105/100 - 1 = 0.05 = 5%. Log return: ln(105/100) = ln(1.05) = 0.04879. If another log return is 0.0100, total log return is 0.05879 and total growth is exp(0.05879)-1.